The limit of a function

Making “approaches without reaching” precise, and the four techniques that evaluate almost every limit you will meet.

LESSON 3–4 | 2 × 40 MIN | ALGEBRA 11, §1.2 | P1 · 7.1

LESSON CLOCK

10' 14' 22' 18' 20' 6'

What a limit says	10 min
Substitution, when it works	14 min
The 0/0 form	22 min
Algebra of limits	18 min
Practice	20 min
Homework	6 min

LEARNING OBJECTIVES

- State informally what lim f(x) = L means.
- Evaluate limits by substitution where the function is continuous.
- Resolve a 0/0 limit by factorising, by rationalising, or by cancelling.
- Use the algebra of limits for sums, products and quotients.

TEXTBOOK

National	pp. 15–26
Cambridge	Pure Mathematics 1, pp. 126–129
Workbook	P1 Exercise 7A

01 Explanation

What the notation says

$$\lim_{x \to a} f(x) = L$$

IN WORDS

As x gets close to a — from either side, but never equal to a — the value $f(x)$ gets and stays as close to L as you like.

The value $f(a)$ is **irrelevant**. The function need not even be defined there. $\frac{x^2 - 1}{x - 1}$ has no value at $x = 1$, yet its limit there is 2: everywhere except at 1 it equals $x + 1$.

Substitution, and when it fails

For a polynomial, or any function continuous at a, the limit is just the value:

$$\lim_{x \to 2} (x^2 + 3x) = 4 + 6 = 10$$

Substitution fails when it produces an **indeterminate form** — most often $\frac{0}{0}$. That is not an answer; it is a signal that algebra is needed first.

> $\frac{0}{0}$ IS NOT ZERO AND NOT ONE
>
> It means the numerator and denominator both vanish, and their **rate** of vanishing decides the answer. Different pairs give different limits.

Three techniques for $\frac{0}{0}$

SHAPE	TECHNIQUE	EXAMPLE
polynomial over polynomial	factorise and cancel	$\frac{x^2 - 9}{x - 3} \to x + 3 \to 6$
a square root appears	multiply by the conjugate	$\frac{\sqrt{x} - 2}{x - 4} \to \frac{1}{\sqrt{x} + 2} \to \frac{1}{4}$
a compound fraction	combine, then cancel	$\frac{\frac{1}{x} - \frac{1}{2}}{x - 2} \to -\frac{1}{4}$

All three do the same thing: remove the factor that is making both parts vanish.

$$\frac{x^2 - 9}{x - 3} = \frac{(x - 3)(x + 3)}{x - 3} = x + 3 \ (x \neq 3)$$

The cancellation is legal precisely because $x \neq 3$ throughout the limiting process.

The algebra of limits

If both limits exist, limits pass through the four operations:

$$\lim(f \pm g) = \lim f \pm \lim g \cdot \lim(f \cdot g) = \lim f \cdot \lim g \cdot \lim \frac{f}{g} = \frac{\lim f}{\lim g} \text{ (if } \lim g \neq 0)$$

and $\lim(c \cdot f) = c \cdot \lim f$. These rules are what let you break a complicated limit into pieces — and the condition on the quotient rule is exactly where the $\frac{0}{0}$ cases hide.

WHY THIS LESSON EXISTS

The derivative is defined as one particular limit, of exactly the $\frac{0}{0}$ kind. Every technique here is used again in the next lesson.

02 Worked examples

EXAMPLE 1 **Evaluate $\lim_{x \to 3} \frac{x^2 - x - 6}{x - 3}$.**

1. Substitution gives $\frac{0}{0}$.

   Factorise.

2. $x^2 - x - 6 = (x - 3)(x + 2)$

3. $= x + 2$ for $x \neq 3$.

4. $\to 5$

ANSWER

5

EXAMPLE 2 **Evaluate $\lim_{x \to 0} \dfrac{\sqrt{x+4} - 2}{x}$.**

1. $\dfrac{0}{0}$ — multiply by the conjugate.

2. $\dfrac{(x+4) - 4}{x(\sqrt{x+4} + 2)}$

3. $= \dfrac{1}{\sqrt{x+4} + 2}$

4. $\to \dfrac{1}{4}$

ANSWER

$\dfrac{1}{4}$

EXAMPLE 3 **Evaluate $\lim_{x \to 2} \dfrac{x^3 - 8}{x^2 - 4}$.**

1. $x^3 - 8 = (x-2)(x^2 + 2x + 4)$

2. $x^2 - 4 = (x-2)(x+2)$

3. $= \dfrac{x^2 + 2x + 4}{x + 2}$

4. $\to \dfrac{12}{4} = 3$

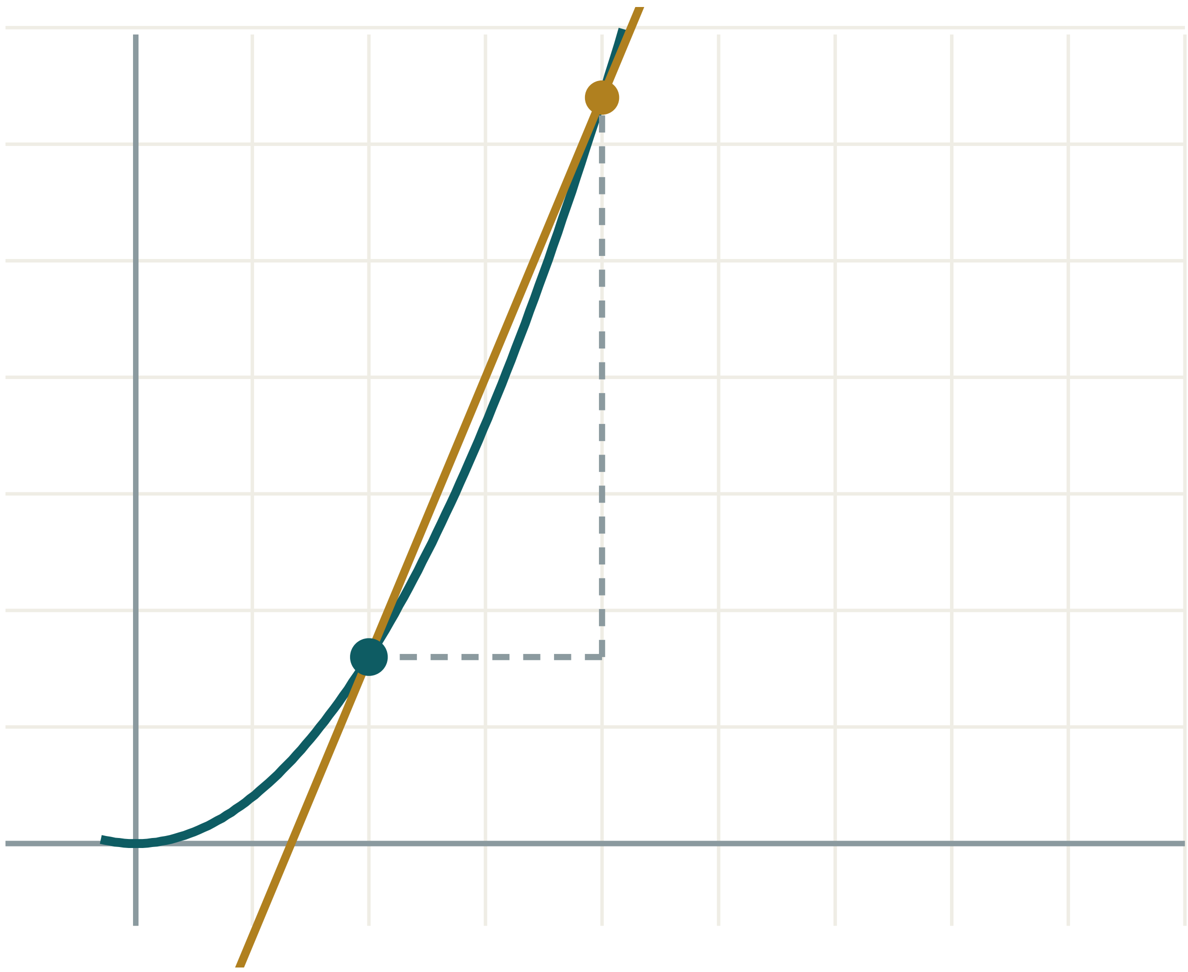
ANSWER

3

03 Interactive model

Tabulate the function either side of the point before doing any algebra.

Watching a limit



x **2** | h **2** | Δy / Δx **2.4** | f ′(x) exactly **1.6**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Limit	Limit	Предел
Tends to	Intiladi	Стремится к
Continuous function	Uzluksiz funksiya	Непрерывная функция
Point of discontinuity	Uzilish nuqtasi	Точка разрыва
Indeterminate form	Aniqmaslik	Неопределённость
One-sided limit	Bir tomonlama limit	Односторонний предел
Removable discontinuity	Bartaraf etiluvchi uzilish	Устранимый разрыв
Conjugate expression	Qo‘shma ifoda	Сопряжённое выражение
Infinity	Cheksizlik	Бесконечность

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

$\lim_{x \to 2} (x^2 + 1)$ is:

- 4
- 5
- 3
- undefined

$\frac{0}{0}$ means:

- the limit is 0
- the limit is 1
- more work is needed
- the limit does not exist

$\lim_{x \to 1} \frac{x^2 - 1}{x - 1}$ is:

- 0
- 1
- 2
- undefined

To handle $\frac{\sqrt{x} - 3}{x - 9}$ you should:

- substitute
- factorise
- multiply by the conjugate
- give up

The value $f(a)$ affects $lim_{x \to a} f(x)$:

- always
- never
- sometimes
- only if defined

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. $\lim_{x \to 1} (3x + 2)$

   ANSWER 5

2. $\lim_{x \to 0} (x^2 - 4)$

   ANSWER -4

3. $\lim_{x \to 2} \frac{x^2 - 4}{x - 2}$

   ANSWER 4

4. $\lim_{x \to 3} \frac{x^2 - 9}{x - 3}$

   ANSWER 6

5. $\lim_{x \to 5} 7$

   ANSWER 7

6. $\lim_{x \to 1} \frac{x - 1}{x^2 - 1}$

   ANSWER 0.5

7. Is $\frac{0}{0}$ an answer?

   ANSWER no

Medium · the standard question

7 PROBLEMS

1. $$\lim_{x \to -2} \frac{x^2 + 3x + 2}{x + 2}$$

ANSWER -1

2. $$\lim_{x \to 4} \frac{\sqrt{x} - 2}{x - 4}$$

ANSWER $\frac{1}{4}$

3. $$\lim_{x \to 0} \frac{\sqrt{x + 9} - 3}{x}$$

ANSWER $\frac{1}{6}$

4. $$\lim_{x \to 2} \frac{x^3 - 8}{x - 2}$$

ANSWER 12

5. $$\lim_{x \to 1} \frac{x^3 - 1}{x^2 - 1}$$

ANSWER 1.5

6. $$\lim_{x \to 3} \frac{\frac{1}{x} - \frac{1}{3}}{x - 3}$$

ANSWER $-\frac{1}{9}$

7. $$\lim_{x \to 0} \frac{(2 + x)^2 - 4}{x}$$

ANSWER 4

Hard · stretch and reason

7 PROBLEMS

1. $\lim_{x \to 1} \dfrac{x^4 - 1}{x - 1}$

ANSWER 4

2. $\lim_{x \to 0} \dfrac{\sqrt{1 + x} - \sqrt{1 - x}}{x}$

ANSWER 1

3. $\lim_{x \to 2} \dfrac{x^3 - 3x^2 + 4}{x^2 - 4}$

ANSWER $\dfrac{3}{4}$

4. $\lim_{x \to a} \dfrac{x^2 - a^2}{x - a}$

ANSWER $2a$

5. $\lim_{x \to a} \dfrac{x^n - a^n}{x - a}$

ANSWER na^{n-1}

6. Find k so that $\dfrac{x^2 + kx - 6}{x - 2}$ has a finite limit at 2

ANSWER $k = 1$

7. Give two functions with $\dfrac{0}{0}$ at 0 and different limits

ANSWER $\dfrac{x}{x} \to 1$ and $\dfrac{x^2}{x} \to 0$

07 Homework

Homework — 5 tasks

Show the algebra. A limit written down with no working scores nothing.

1. Evaluate $\lim_{x \to 4} \frac{x^2 - 16}{x - 4}$.
2. Evaluate $\lim_{x \to 0} \frac{\sqrt{x + 16} - 4}{x}$.
3. Evaluate $\lim_{x \to 3} \frac{x^3 - 27}{x^2 - 9}$.
4. Explain in three sentences why $\frac{0}{0}$ is called indeterminate.
5. Evaluate $\lim_{x \to a} \frac{x^3 - a^3}{x - a}$ and check it against na^{n-1}.